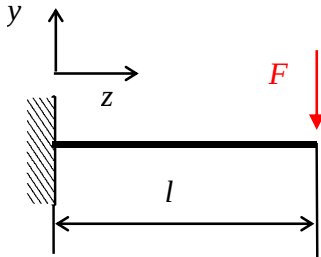


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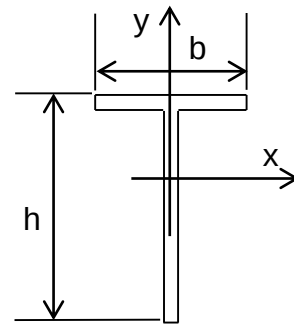
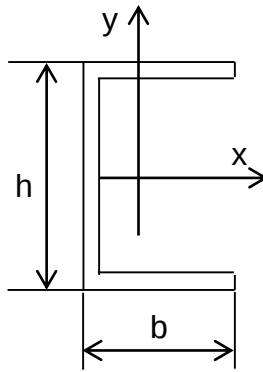
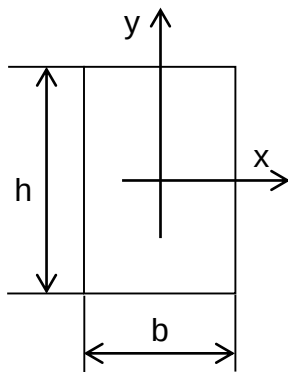
Tutorial #8: Shear forces and buckling phenomenon

Example 1

A cantilever beam is loaded at its tip by a force, F . Find first the most stressed cross section.



For each cross section herein drawn, show the distribution of bending and shear stresses, respectively. Find the point where stress achieves its maximum value for bending then for shear. Assume that: $h = 30$ mm, $b = 20$ mm, $t = 2$ mm, $l = 1$ m, $F = 500$ N.



[rectangle: $\sigma_{zz,\max} = \pm 167$ MPa, $\tau_{yz,\max} = -1.3$ MPa

C-shape: $\sigma_{zz,\max} = \pm 402$ MPa, $\tau_{yz,\max} = -10$ MPa

T-Shape: $\sigma_{zz,\min} = -1135$ MPa, $\sigma_{zz,\max} = 546$ MPa, $\tau_{yz,\max} = -12$ MPa]

Example 2

For the C-section of above example 1, compute the shear stress due to vertical and horizontal shear forces $F_y = 250$ N and $F_x = 250$ N applied to the centroid of the cross section. Furthermore, evaluate the shear stress due to the torsional moment induced by F_y .

$[x_e = -13$ mm, $y_e = 0]$

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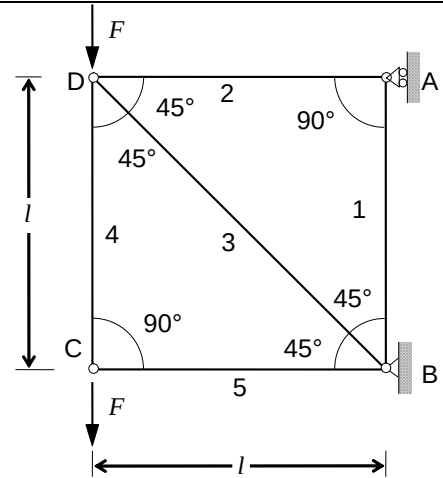
Tutorial #8: Shear forces and buckling phenomenon

Example 3

For the truss structure ($l = 2000$ mm) shown in figure, compute the safety factor of beams against yielding and buckling, respectively, being properties of material: $R_{eH} = \sigma_{YY} = 230$ MPa, $E = 2 \times 10^5$ MPa, $\nu = 0.3$. Rods have circular and hollow section, with outer diameter $D = 50$ mm and inner diameter $d = 40$ mm.

$F = 20$ kN

[Yielding: $F_{S,min} = 2.9$; Buckling: $F_S < 1$]



Shear centre: an experiment.

http://www.mace.manchester.ac.uk/project/teaching/civil/structuralconcepts/Statics/shear/shear_mod6.php

